

## Supplemental Material

### Acute Cardiovascular Effects of Controlled Exposure to Dilute Petrodiesel and Biodiesel Exhaust Emissions in Healthy Volunteers: A Crossover Study

**Table S1**

Total metal and metalloid concentrations measured in PM<sub><0.2</sub> and PM<sub>0.2-0.5</sub> fractions derived from petrodiesel and RME blended fuels.

| Element<br>(µg/mg) | Petrol diesel         |                       | RME30                 |                       | RME100                |                       |
|--------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                    | PM <sub>&lt;0.2</sub> | PM <sub>0.2-0.5</sub> | PM <sub>&lt;0.2</sub> | PM <sub>0.2-0.5</sub> | PM <sub>&lt;0.2</sub> | PM <sub>0.2-0.5</sub> |
| <sup>27</sup> Al   | 0.38±0.36             | 0.70±1.19             | 1.35±2.33             | 0.08±0.13             | 7.56±10.18            | 0.19±0.33             |
| <sup>75</sup> As   | 0.01±0.01             | 0.05±0.01             | 0.00±0.00             | <b>0.00±0.00</b>      | 0.00±0.00             | 0.05±0.05             |
| <sup>135</sup> Ba  | 0.03±0.02             | 0.04±0.04             | 0.03±0.04             | 0.03±0.02             | 0.06±0.03             | 0.02±0.02             |
| <sup>55</sup> Mn   | 0.23±0.22             | 0.47±0.30             | 2.31±2.49             | <b>3.96±2.14</b>      | 2.23±2.95             | <b>6.98±3.59</b>      |
| <sup>50</sup> V    | 1.15±0.99             | 4.57±3.92             | 27.00±28.56           | <b>47.82±23.10</b>    | 26.39±37.60           | <b>89.21±48.50</b>    |
| <sup>52</sup> Cr   | 1.23±1.08             | 5.01±4.31             | 29.47±31.59           | <b>50.82±25.40</b>    | 28.61±40.67           | <b>96.35±52.97</b>    |
| <sup>63</sup> Cu   | 0.25±0.19             | 0.81±0.59             | 0.24±0.17             | <b>0.38±0.12</b>      | 0.68±0.53             | 0.77±0.41             |
| <sup>95</sup> Mo   | 0.02±0.01             | 0.05±0.03             | 0.33±0.35             | <b>0.63±0.35</b>      | 0.28±0.40             | <b>0.92±0.50</b>      |
| <sup>60</sup> Ni   | 1.09±0.73             | 3.00±2.43             | 15.07±16.51           | <b>25.12±12.86</b>    | 14.52±20.31           | <b>48.03±25.89</b>    |
| <sup>66</sup> Zn   | 2.61±0.48             | 4.83±2.08             | 3.47±0.66             | 2.25±0.69             | 2.48±1.23             | 3.04±0.62             |
| <sup>44</sup> Ca   | 5.71±1.19             | 13.42±5.20            | 2.46±1.24             | 1.79±1.44             | 9.46±8.10             | 10.55±3.48            |
| <sup>56</sup> Fe   | 5.45±4.19             | 19.21±15.00           | 106.51±114.62         | <b>180.76±101.44</b>  | 96.29±131.28          | <b>305.33±145.50</b>  |
| <sup>11</sup> B    | 0.31±0.13             | 0.97±0.88             | <b>0.00±0.00</b>      | 0.00±0.00             | <b>0.07±0.12</b>      | 0.07±0.12             |
| <sup>197</sup> Au  | 0.14±0.16             | 0.40±0.49             | 0.00±0.00             | 0.00±0.00             | 0.02±0.04             | 0.00±0.00             |
| <sup>88</sup> Sr   | 0.02±0.00             | 0.04±0.02             | <b>0.01±0.00</b>      | <b>0.01±0.00</b>      | 0.02±0.01             | 0.02±0.02             |
| <sup>59</sup> Co   | 0.02±0.02             | 0.03±0.02             | 0.20±0.21             | <b>0.35±0.19</b>      | 0.18±0.25             | <b>0.58±0.3</b>       |
| <sup>208</sup> Pb  | 0.03±0.01             | 0.06±0.04             | 0.02±0.01             | 0.01±0.01             | 0.04±0.05             | 0.02±0.01             |
| <sup>85</sup> Rb   | 0.01±0.01             | 0.01±0.01             | 0.00±0.00             | 0.00±0.00             | 0.01±0.01             | 0.01±0.00             |
| (ng/mg)            |                       |                       |                       |                       |                       |                       |
| <sup>9</sup> Be    | 0.00±0.00             | 0.29±0.49             | 0.21±0.19             | 0.35±0.61             | 0.27±0.26             | 0.40±0.69             |
| <sup>121</sup> Sb  | 2.20±1.25             | 1.54±1.49             | 2.07±2.71             | 1.41±1.48             | 2.48±1.70             | 4.82±2.72             |
| <sup>111</sup> Cd  | 0.00±0.00             | 0.00±0.00             | 3.30±5.72             | 2.57±3.93             | 0.00±0.00             | 0.00±0.00             |
| <sup>133</sup> Cs  | 0.09±0.07             | 0.16±0.12             | 0.03±0.05             | 0.00±0.00             | 0.14±0.14             | 0.10±0.04             |

Data represent the mean (SD) of measurements made on PM extracts from three separate PM filters collected during separate exposures. All values are expressed as µg/mg PM, with the exception of Be, Sb, Cd and Cs, which are expressed as ng/mg PM. Figures in bold illustrate significant difference (P<0.05) in concentration within a given fraction relative to Petrodiesel value. Shaded rows illustrate elements where elements were quantified following reaction with ammonia using the dynamic reaction cell of the ICP-MS to remove potential polyatomic interferences.

Table S2

Exposure conditions from earlier studies with bilateral forearm plethysmography after petrodiesel exhaust and filtered air in healthy human subjects.

| Study (ref)              | Mills et al 2005 (23) | Törnqvist et al 2007 (24) | Barath et al 2010 (25) | Lucking et al 2011 (27) |
|--------------------------|-----------------------|---------------------------|------------------------|-------------------------|
| Engine mode              | Idling                | Idling                    | ETC                    | ETC                     |
| PM1 (ug/m <sup>3</sup> ) | 300                   | 300                       | 250                    | 320                     |
| NO <sub>2</sub> (ppm)    | 1.6                   | n/a                       | 0.9                    | 0.7                     |
| NOx (ppm)                | 6.1                   | n/a                       | 7.5                    | 6.4                     |
| THC (ppm)                | 4.3                   | n/a                       | 1.2                    | 0.91                    |
| Number of subjects       | 30                    | 15                        | 18                     | 27                      |

Data given as mean values. ETC=European Transient Cycle, urban part. THC= total hydrocarbons. n/a; data not available but was part of the same exposure series and indicated to be similar to Mills 2005. Combined plethysmography data are presented in Figure S1.

Table S2

## Change in Blood Cell Counts

|  |                                                 | <i>Hours post exposure</i> | <i>Petrodiesel</i>   | <i>RME30</i>        | <i>p value</i>  |             | <i>Petrodiesel</i>   | <i>RME100</i>        | <i>p value</i>  |             |
|--|-------------------------------------------------|----------------------------|----------------------|---------------------|-----------------|-------------|----------------------|----------------------|-----------------|-------------|
|  |                                                 |                            |                      |                     | <i>Exposure</i> | <i>Time</i> |                      |                      | <i>Exposure</i> | <i>Time</i> |
|  | <i>Change in Leukocytes, x10<sup>9</sup>/L</i>  | <b>2 h</b>                 | 0.89 [0.21, 1.56]    | 0.77 [0.01, 1.45]   | 0.67            | <0.01       | 0.56 [0.05, 1.17]    | 0.83 [0.33, 1.33]    | 0.28            | 0.02        |
|  |                                                 | <b>4 h</b>                 | 0.67 [0.10, 1.26]    | 0.77 [0.15, 1.39]   |                 |             | 0.56 [0.01, 1.11]    | 1.00 [0.56, 1.44]    |                 |             |
|  |                                                 | <b>8 h</b>                 | 1.01 [0.31, 1.84]    | 1.27 [0.43, 2.11]   |                 |             | 1.11 [0.31, 1.90]    | 0.62 [0.04, 1.20]    |                 |             |
|  |                                                 | <b>24 h</b>                | 0.04 [-0.37, 0.45]   | 0.01 [-0.50, 0.48]  |                 |             | -0.06 [-0.34, 0.21]  | -0.14 [-0.34, 0.24]  |                 |             |
|  | <i>Change in Lymphocytes, x10<sup>9</sup>/L</i> | <b>2 h</b>                 | -0.29 [-0.48, -0.11] | -0.18 [-0.40, 0.04] | 0.71            | <0.01       | -0.32 [-0.60, -0.04] | -0.29 [-0.54, 0.04]  | 0.45            | 0.61        |
|  |                                                 | <b>4 h</b>                 | -0.27 [-0.53, 0.02]  | -0.23 [-0.43, 0.03] |                 |             | -0.17 [-0.45, 0.10]  | -0.07 [-0.32, 0.19]  |                 |             |
|  |                                                 | <b>8 h</b>                 | 0.03 [-0.22, 0.29]   | 0.07 [-0.09, 0.24]  |                 |             | -0.08 [-0.40, 0.23]  | -0.09 [-0.33, 0.15]  |                 |             |
|  |                                                 | <b>24 h</b>                | 0.14 [-0.10, 0.38]   | 0.09 [-0.10, 0.28]  |                 |             | -0.11 [-0.26, 0.04]  | -0.14 [-0.28, -0.01] |                 |             |
|  | <i>Change in Monocytes, x10<sup>9</sup>/L</i>   | <b>2 h</b>                 | -0.08 [-0.14, 0.01]  | -0.04 [-0.12, 0.04] | 0.22            | 0.02        | 0.01 [-0.04, 0.05]   | 0.04 [-0.01, 0.09]   | 0.07            | 0.41        |
|  |                                                 | <b>4 h</b>                 | -0.05 [-0.14, 0.01]  | -0.02 [-0.07, 0.03] |                 |             | 0.00 [-0.05, 0.5]    | 0.03 [-0.03, 0.08]   |                 |             |
|  |                                                 | <b>8 h</b>                 | 0.01 [-0.08, 0.10]   | 0.03 [-0.07, 0.14]  |                 |             | 0.03 [-0.03, 0.10]   | -0.01 [-0.07, 0.06]  |                 |             |
|  |                                                 | <b>24 h</b>                | 0.03 [-0.10, 0.05]   | 0.01 [-0.05, 0.06]  |                 |             | -0.11 [-0.26, 0.04]  | -0.02 [-0.07, 0.03]  |                 |             |
|  | <i>Change in Neutrophils, x10<sup>9</sup>/L</i> | <b>2 h</b>                 | 1.29 [0.73, 1.85]    | 1.04 [0.48, 1.60]   | 0.63            | <0.01       | 0.89 [0.38, 1.41]    | 1.13 [0.62, 1.64]    | 0.27            | <0.01       |
|  |                                                 | <b>4 h</b>                 | 1.05 [0.61, 1.49]    | 1.10 [0.49, 1.71]   |                 |             | 0.76 [0.38, 1.13]    | 1.11 [0.73, 1.49]    |                 |             |
|  |                                                 | <b>8 h</b>                 | 1.09 [0.48, 1.70]    | 1.30 [0.37, 2.23]   |                 |             | 1.19 [0.60, 1.78]    | 0.76 [0.40, 1.11]    |                 |             |
|  |                                                 | <b>24 h</b>                | -0.10 [-0.39, 0.19]  | -0.05 [-0.43, 0.32] |                 |             | 0.07 [-0.21, 0.36]   | 0.14 [-0.03, 0.32]   |                 |             |
|  | <i>Change in platelets, x10<sup>9</sup>/L</i>   | <b>2 h</b>                 | -11 [-19, -4]        | -15 [-21, -9]       | 0.09            | <0.01       | -11 [-17, -7]        | -9 [-16, -3]         | 0.67            | <0.01       |
|  |                                                 | <b>4 h</b>                 | -10 [-15, -4]        | -16 [-21, -11]      |                 |             | -12 [-18, -7]        | -8 [-17, 0]          |                 |             |
|  |                                                 | <b>8 h</b>                 | -19 [-30, -8]        | -18 [-25, -11]      |                 |             | -21 [-28, -15]       | -24 [-39, -10]       |                 |             |
|  |                                                 | <b>24 h</b>                | 3 [-5, 10]           | -3 [-13, 7]         |                 |             | 8 [2, 14]            | 8 [1, 16]            |                 |             |

Absolute change in number of cells x10<sup>9</sup>/L in peripheral blood, compared to pre-exposure, mean with 95% CI. Left panel Petrodiesel vs. RME30 n=16. Right panel Petrodiesel vs. RME100 n=19. Changes in blood cell counts were similar after either exposure with a transient increase in leukocytes and neutrophils along with a reduction of platelets. All values returned to pre-exposure levels over 24 hours. P values for type of exposure and change over time from 2-way repeated measures ANOVA

44 **Table S3**45 **Platelet activation**

|                                                          | <i>Hours<br/>post<br/>expos<br/>ure</i> | <i>Petrodiesel</i>   | <i>RME30</i>         | <i>P</i> | <i>Petrodiesel</i>   | <i>RME100</i>        | <i>P</i> |
|----------------------------------------------------------|-----------------------------------------|----------------------|----------------------|----------|----------------------|----------------------|----------|
| <b><i>Platelet<br/>Monocyte<br/>Binding, %</i></b>       | 2                                       | 35.5<br>[30.0, 40.9] | 38.5<br>[31.7, 45.3] | 0.43     | 36.4<br>[25.9, 46.9] | 28.9<br>[20.3, 37.4] | 0.15     |
|                                                          | 4                                       | 42.4<br>[36.3, 48.5] | 39.9<br>[33.2, 46.6] | 0.53     | 31.4<br>[22.7, 40.2] | 27.8<br>[19.6, 35.9] | 0.53     |
| <b><i>Monocyte<br/>CD40 positivity,<br/>%</i></b>        | 2                                       | 18.3<br>[18.3, 19.9] | 19.9<br>[12.8, 27.0] | 0.53     | 3.8<br>[1.6, 6.0]    | 2.3<br>[0.6, 3.9]    | 0.25     |
|                                                          | 4                                       | 18.7<br>[11.6, 25.9] | 17.4<br>[12.6, 22.2] | 0.67     | 2.8<br>[1.3, 4.4]    | 2.3<br>[1.2, 3.3]    | 0.46     |
| <b><i>Platelet CD40<br/>Ligand<br/>Positivity, %</i></b> | 2                                       | 0.5<br>[0.4, 0.6]    | 0.5<br>[0.3, 0.6]    | 0.73     | 0.3<br>[0.0, 0.5]    | 0.2<br>[0.1, 0.2]    | 0.45     |
|                                                          | 4                                       | 0.4<br>[0.3, 0.5]    | 0.5<br>[0.3, 0.6]    | 0.81     | 0.2<br>[0.1, 0.3]    | 0.2<br>[0.1, 0.3]    | 0.93     |
| <b><i>Platelet p-<br/>selectin<br/>expression, %</i></b> | 2                                       | 1.0<br>[0.8, 1.3]    | 1.4<br>[0.8, 2.1]    | 0.18     | 0.5<br>[0.2, 0.8]    | 0.4<br>[0.1, 0.6]    | 0.31     |
|                                                          | 4                                       | 1.3<br>[0.8, 1.7]    | 1.0<br>[0.7, 1.2]    | 0.11     | 0.3<br>[0.2, 0.4]    | 0.4<br>[0.1, 0.6]    | 0.60     |

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47 Platelet activation assessed with flow cytometry was similar after either exposure. Left panel

48 Petrodiesel vs. RME30 n=16. Right panel Petrodiesel vs. RME100 n=19. P values from

49 Student's t-test, mean with 95% CI.

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51 **Table S4**

52 **Fraction of exhaled NO**

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|                            |                    | Study 1           |                  |    |      | Study 2           |                   |    |      |
|----------------------------|--------------------|-------------------|------------------|----|------|-------------------|-------------------|----|------|
|                            |                    | Petrodiesel       | RME30            | n  | p    | Petrodiesel       | RME100            | n  | p    |
| FE <sub>NO</sub> 50<br>ppb | ml/s, Pre exposure | 12.0 [8.9, 15.2]  | 11.1 [8.5, 13.7] | 11 |      | 12.9 [9.9, 16.0]  | 15.3 [11.2, 19.3] | 18 |      |
|                            | 8 h post exposure  | 14.7 [10.1, 19.4] | 13.3 [9.6, 17.1] | 13 | 0.25 | 15.1 [11.4, 18.8] | 14.6 [10.6, 18.6] | 16 | 0.70 |

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56 Fraction of exhaled NO at 50 ml/s, ppb. Mean values with 95% CI, p value for paired t-test 8 h post exposure.

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59 **Figure S1 (also given in separate file)**

60 **Legend:**

61 Combined plethysmography data from four studies comparing the effects of petrodiesel with filtered air (ref 23, 24, 25, 27). Petrodiesel consistently  
 62 causes vasomotor dysfunction to endothelial (acetylcholine, bradykinin, nitroprusside) and non-endothelial dependent vasodilators (verapamil) [vs.](#)  
 63 [filtered air](#). Technical and sampling difficulties during bilateral [forearm brachial](#) arterial needle infusions of vasoactive agents were reasons for not  
 64 obtaining full sets of data from all individuals.

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